

Hossam Khalil

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Education

PhD in Astrophysics, University of Helsinki (UH), Finland Jan 2025 -Present

Working Title: *Improving cosmological constraints from large-scale structure in the era of eROSITA and Euclid*

Supervisor: Alexis Finoguenov

M.Sc. in Astrophysics, University of Helsinki (UH), Finland Sep 2022 -Jun 2024

Title: *AXES-2MRS: A new all-sky catalogue of extended X-ray galaxy groups*

Supervisor: Alexis Finoguenov

B.Sc. in Physics, University of Science & Technology at Zewail City (UST), Egypt Sep 2015 - Jul 2019

Grants & Fellowships

The Vilho, Yrjö and Kalle Väisälä Foundation of the Finnish Academy of Science and Letters Grant
(EUR 13500 six-month working grant. Declined in favor of a PhD employment contract) Jan 2025

Research Experience

Master Thesis Jun 2024

Department of Physics at University of Helsinki (Supervisor: Alexis Finoguenov)

Helsinki, Finland

- Released a new all-sky catalogue of X-ray detected groups (AXES-2MRS) by identifying extended X-ray sources in the ROSAT all-sky survey with the 2MRS optical group catalogue. Performed X-ray analysis on a subset observed by XMM-Newton to study the baryonic properties of these groups through X-ray-optical scaling relations.

Research Trainee in Gravitational Waves Sep 2023 - Jan 2024

Helsinki Institute of Physics & Department of Physics at UH (Supervisor: David J. Weir)

Helsinki, Finland

- Analyzed the NANOGrav 15-year data set to constrain the parameters of a stochastic gravitational wave background signal from a first-order phase transition using PTArcade.

Nordic Optical Telescope (NOT) Remote Observing School Oct 2022

University of Helsinki and University of Turku

Turku, Finland

- Conducted optical observations of nearby galaxy groups using the NOT in La Palma.
- Carried out the observation preparation, spectral reduction, and redshift analysis to complete cluster surveys.

Zwicky Transient Facility (ZTF) Summer School Jul 2022

University of Minnesota and Caltech

online

- Performed multi-messenger astrophysics analysis tasks including transient detection in ZTF datasets and optical follow-ups to neutrino (IceCube) and gravitational wave (LIGO & VIRGO) alerts.

Research Trainee in High Energy Astrophysics Aug 2017 - Jan 2018

UST at Zewail City (Supervisor: Alaa Ibrahim)

Giza, Egypt

- Studied different types of bursts from a radio-emitting magnetar and performed spectral fitting to model the emission region properties.

Publications

- H. Khalil**, A. Finoguenov, E. Tempel, and G. A. Mamon, *AXES-2MRS: A new all-sky catalogue of extended X-ray galaxy groups*, 2024, A&A, 690, A212 [[arXiv](#)].

Professional Experience

Physics Teaching Assistant
UST at Zewail City

Sep 2020 - Jun 2022
Giza, Egypt

- Supervised 50+ students in PEU 327 (Observational Astrophysics Lab) and PEU 450 (Computational Physics).
- Delivered online and in-person lectures, graded assignments, and provided one-on-one support clinics.

Data Analyst
EV Mobility (Mansour Group)

Jan 2020 - Jan 2021
Cairo, Egypt

- Worked in the R&D department, contributing to a project that aimed at introducing electric micromobility solutions to Egypt.
- Analyzed the global electric micromobility scene, conducted market analysis, and provided recommendations that had high-level business consequences.

Scientific Language Editing Trainee
Natural Sciences Publishing

Aug 2019 - Jan 2020
Giza, Egypt

- Worked as a trainee in the Applied Mathematics & Information Sciences Journal and Journal of Statistics Applications & Probability.

Invited Talks

Max Planck Institute for Astronomy
GC Theory Group

14 Nov 2024
Heidelberg, Germany (*online*)

XMM-Newton Group AGN Project
X-GAP Collaboration

18 Jun 2024
Versoix, Switzerland (*online*)

University of Rome Tor Vergata
Department of Physics

29 Jan 2024
Rome, Italy

Academic Achievements

University of Helsinki Scholarship Covering 100% of my total master's program fees.

Zewail City Excellence Scholarship Covering 80% of my total bachelor's program fees.

Provost's Honors Rol Fall 15', Spring 16'.

Skills

Astrophysics Software

SAS, eSASS, HEASoft, PTArcade, Xspec, IRAF, CARTA, TOPCAT

General Programming Languages

Python, Mathematica, MATLAB

Other Software

MS Office, Adobe Photoshop

Languages

Arabic (Native speaker), English (Fluent), Finnish (Beginner)

Extracurricular Activities

- Vice president and active member at Zewail City Astronomy Club — (2016-2018).
- Co-founder, vice president, and tutor at Zewail City Chess Club — (2016-2019).
- Won the Zewail City Student Union Award for creative writing — (September 2016).

REFERENCES ARE AVAILABLE UPON REQUEST